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Multi-Condition Diagnosis Framework and Dual-Model Fusion Diagnosis Algorithm for Network Faults

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Abstract

As high-performance computing expands from large-scale data analytics to deep learning, multiprocessor systems are becoming increasingly complex and widely deployed across industrial automation, intelligent manufacturing, cloud infrastructure, and big data processing. Timely fault diagnosis is therefore essential for maintaining system reliability and operational efficiency. Multi-condition diagnostic strategies combined with efficient detection algorithms can identify and isolate faults early, thereby reducing potential economic losses. In this study, we investigate multi-condition diagnosis and derive the 1-extra r-component diagnosability of the half hypercube network HH_n under the MM* model. We then integrate the MM* and PMC models to develop a dual-model fusion diagnosis algorithm, termed $DMFDA$, and evaluate it through simulations on HH_n. Results show that $DMFDA$ achieves strong diagnostic performance: at a 30% fault rate, the TPR remains above 0.9951, while the FPR does not exceed 0.0031. We further compare $DMFDA$ with other algorithms (ICFD-P and ACP-FD) through fault injection simulations on the $DCell_{(2,4)}$ topology. At a 50% fault rate, the mean TPR values of $DMFDA$, ICFD-P, and ACP-FD are 0.9925, 0.8337, and 0.9336, respectively, while their mean FPR values are 0.1267, 0.0649, and 0.0876, respectively.

Key words

Fault diagnosis; Multi-condition diagnosability; MM* model; Fusion diagnosis algorithm

1 Introduction

The proliferation of high-performance computing applications, ranging from large-scale data analytics to deep learning workloads, has driven the widespread domain of increasingly complex multiprocessor systems. As the count of processing elements within such architectures expands, the likelihood of processor failures rises accordingly, posing a direct threat to system dependability. Ensuring operational continuity thus necessitates the timely detection and isolation of faulty processors—a task formally referred to as system-level diagnosis [1]. This discipline has gained considerable traction across a variety of domains, including cloud computing infrastructures [2], smart grid infrastructures [3,4], and data center networks [5].

System-level diagnosis, a key technique in fault management, identifies faulty processors by analyzing test results, thereby allowing for their prompt replacement. So far, the field has mainly focused on two distinct diagnostic approaches: the PMC model, originally proposed by Preparata et al. [1], and the MM* model, introduced by Sengupta and Dahbura [6]. The PMC model is a test-based diagnostic framework, where any vertex can test its neighbors via communication links to determine whether they are faulty. In contrast, the MM* model adopts a comparison-based approach, wherein a vertex assigns a common task to a pair of neighboring vertices and then compares their responses. The consistency or inconsistency of these responses serves as the basis for identifying the set of faulty vertices. If a system can successfully identify every faulty unit under the premise that at most t processors are faulty, then the system is defined as t-diagnosable. A network's diagnosability reflects its ability to withstand faults, which manifests as the capacity to accurately and quickly locate faulty vertices and carry out necessary replacements.

To strengthen the fault tolerance and robustness of multiprocessor systems, various reliability metrics have been progressively proposed, including extra connectivity [12], extra diagnosability [13], g-good-neighbor diagnosability [14], component diagnosability [15]. In recent years, the aforementioned reliability measures have gained widespread academic interest. Yin and Xu [16] and Guo et al. [17] investigated the h-extra connectivity of different network families: the former focused on enhanced hypercubes, while the latter systematically studied the h-extra connectivity of folded crossed cubes. Wang [18] calculated the h-extra diagnosability of the Möbius cube under the PMC and MM* models, while Li et al. [19] provided the 3-extra diagnosability of the balanced hypercube under the MM* model. Regarding g-good-neighbor diagnosability, Asgharian-Sardroud and Ghasemi [20]

Table 1 A summary of different reliability metrics for the half hypercube network HH_n.

Reliability metric	Theoretical result	Condition	Model	Reference
h-extra connectivity	$\kappa_h(HH_n) = -\frac{1}{2}h^2 + \left(\lceil\frac{n}{2}\rceil - \frac{1}{2}\right)h + \lceil\frac{n}{2}\rceil + 1$	$n \geq 7, 0 \leq h \leq \lceil\frac{n}{2}\rceil - 4$	—	[7]
h-extra diagnosability	$t_h(HH_n) = -\frac{1}{2}h^2 + \left(\lceil\frac{n}{2}\rceil + \frac{1}{2}\right)h + \lceil\frac{n}{2}\rceil + 1$	$n \geq 7, 0 \leq h \leq \lceil\frac{n}{2}\rceil - 4$	PMC	[7]
		$n \geq 7, 0 \leq h \leq \frac{1}{4}\lceil\frac{n}{2}\rceil - 1$	MM*	[7]
g-good-neighbor diagnosability	$t_g(HH_n) = 2^g(\lceil\frac{n}{2}\rceil - g + 2) - 1$	$n \geq 7, 1 \leq g \leq \lceil\frac{n}{2}\rceil - 2$	PMC/MM*	[8]
r-component diagnosability	$t_{r+1}(HH_n) = (r+1)\lceil\frac{n}{2}\rceil - \frac{r(r+1)}{2} + 1$	$1 \leq r \leq \lceil\frac{n}{2}\rceil$	PMC/MM*	[9]
1-extra r-component connectivity	$c\kappa_3^1(HH_n) = 4\lceil\frac{n}{2}\rceil - 4$	$r = 2, n \geq 7$	—	[10]
	$c\kappa_4^1(HH_n) = 6\lceil\frac{n}{2}\rceil - 10$	$r = 3, n \geq 11$	—	[10]
1-extra r-component diagnosability	$ct_2^1(HH_n) = 4\lceil\frac{n}{2}\rceil - 3$	$r = 2, n \geq 13$	PMC	[11]
	$ct_3^1(HH_n) = 6\lfloor\frac{n}{2}\rfloor - 9$	$r = 3, n \geq 23$	PMC	[11]
	Unknown	Unknown	MM*	Unknown

investigated this property for triangle-free graphs under the PMC and MM* models. Meanwhile, Geng et al. [21] determined the 1-good-neighbor diagnosability of a class of folded hypercubes under the same two models. Huang et al. [22] and Zhuo et al. [23] conducted research on the component diagnosability of different networks. Specifically, the former analyzed the r-component diagnosability of hierarchical cubic networks using the PMC and MM* models, while the latter obtained the $\{2, 3\}$-component diagnosability of alternating group graphs under the same two models. Beyond the works discussed above, further studies on single-condition reliability metrics can be found in [24–29]

The above-mentioned investigations are largely limited to single-condition diagnostic frameworks. However, as network size and complexity increase, a single indicator is insufficient to effectively characterize system fault tolerance and robustness. To overcome this limitation, Li et al. [30] defined the h-extra r-component connectivity, requiring that after deleting a faulty set, the surviving network not only contains at least r components but also ensures that each component meets a specified threshold. Under this framework, they computed the 1-extra r-component connectivity of the hypercube network. Later, Jagadale et al. [31] generalized this multi-conditional connectivity to 3-ary n-cube networks. Inspired by this concept, Liao et al. [32] further introduced h-extra r-component diagnosability and determined the theoretical value of this metric for the bubble-sort network under the PMC model. In addition, they proposed a FAFD-PMC diagnostic algorithm to detect and locate faulty vertices in the system. After that, this metric was successively extended to half hypercube [11], star graph [33], and hypercube [34], and their 1-extra $\{2, 3\}$-component diagnosability under the PMC model was obtained. Meanwhile, different diagnostic algorithms—IBDA, TSFD, and ECD-PMC—were proposed to identify faulty processors in the corresponding networks.

Notably, although existing studies have examined the multi-conditional diagnosability of different networks, they have all adopted the PMC diagnostic framework. For the MM* model, the theoretical values of multi-conditional diagnosability remain largely unexplored. The difficulty stems from the fact that isolated vertices must be accounted for in the lower-bound derivation, making the analysis significantly more involved than under the PMC model. To address this research gap, we select the half hypercube as a representative network. This network is widely studied because it maintains the key advantages of hypercubes while cutting vertex degree and network diameter by approximately half. Meanwhile, we summarize the research results of the aforementioned reliability metrics for the half hypercube network in Table 1. Consequently, this paper specifically investigates the 1-extra r-component diagnosability of the half hypercube under the MM* model. The contributions presented in this paper are as follows:

- We derive the precise expressions of the 1-extra r-component diagnosability for the half hypercube network HH_n under the MM* model. Specifically, $ct_2^1(HH_n) = 3\lceil\frac{n}{2}\rceil - 2$ for $n \geq 13$ and $r = 2$; $ct_3^1(HH_n) = 5\lceil\frac{n}{2}\rceil - 6$ for $n \geq 19$ and $r = 3$.
- We design a dual-model fusion diagnosis algorithm ($DMFDA$) by integrating the MM* and PMC models, aiming to detect faulty processors within general networks. The algorithm first employs the testing rules of the MM* framework to screen out a portion of non-faulty processors. Then, under the PMC model, it uses these non-faulty processors as testers to determine the actual status of the other processors.
- We conduct ablation experiments on HH_{13} to compare $DMFDA$ with PMC-only and MM*-only baselines, and perform sensitivity analysis on the number of repeated MM* comparisons. We also conduct fault injection simulations on the $DCell_{(2,4)}$ topology to compare $DMFDA$ with ICFD-P and ACP-FD at fault rates of 30% and 50%. At the 50% fault rate, $DMFDA$ achieves a mean TPR of 0.9925 and a mean FPR of 0.1267.

The remainder of this work is organized as described below. Section 2 presents the terminology and basic concepts employed in this paper; Section 3 establishes the 1-extra $\{2, 3\}$-component diagnosability of HH_n under the MM* diagnostic framework. Section 4 introduces a dual-model fusion diagnosis algorithm ($DMFDA$) that leverages both the MM* and PMC models, conducts comprehensive simulation experiments on HH_n and the $DCell_{(2,4)}$ topology; Section 5 concludes this paper.

Table 2 Main notations.

Notation	Description
$G = (V(G), E(G))$	A connected graph G, with vertex set $V(G)$ and edge set $E(G)$, is used to model an interconnection network
$\|V(G)\|$	The number of vertices in G
(t_1, t_2)	There is an edge between t_1 and t_2 in G
P_k	A k-path P_k consists of k vertices connected in a line by $k-1$ edges
$N_G(t)$	The set of neighbors of t in G
$N_G[t]$	$N_G(t) \cup \{t\}$
$\langle y, z\rangle_x$	The vertices y and z are a pair of neighbors of x in G
Δ	$max\{\|N_G(t)\| : t \in V(G)\}$
$X \subseteq G$	A graph X is said to be a subgraph of G if $V(X) \subseteq V(G)$ and $E(X) \subseteq E(G)$
$G - X$	The subgraph obtained by deleting all vertices in X and and their incident edges from G
$N_G(X)$	$(\bigcup_{t\in X} N_G(t)) - X$
$N_G[X]$	$N_G(X) \cup X$
$\mathcal{Z}_1 \Delta \mathcal{Z}_2$	The set $(\mathcal{Z}_1 \cup \mathcal{Z}_2) \setminus (\mathcal{Z}_1 \cap \mathcal{Z}_2)$
$\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}$	The set of elements not in $\mathcal{Z}_1 \cup \mathcal{Z}_2$
$E[X_1, X_2]$	The edge set between between X_1 and X_2 in G

2 Preliminary

The key terms and formal definitions that underpin the remainder of this work are presented in this section.

2.1 Notations and Concepts

For convenience, the principal notations used in this paper are summarized in Table 2. Next, we present the testing rules of the PMC and MM* diagnostic frameworks, which are as follows.

Under the PMC diagnostic framework, a diagnostic test is performed by a processor x on an adjacent processor y. The outcome of this test is denoted by $\rho(x, y)$. When the tester x is non-faulty, the result is reliable: $\rho(x, y) = 0$ if y is non-faulty, and $\rho(x, y) = 1$ if y is faulty. Conversely, if the tester x is faulty, the test result is arbitrary, meaning $\rho(x, y)$ may be either 0 or 1 regardless of the actual status of y, yielding an unreliable outcome.

Under the MM* diagnostic framework, a vertex w distributes an identical task to two of its neighbors, denoted u and v. Upon receiving the results, w compares the outputs from u and v. The outcomes of such comparisons across the system are collectively referred to as the syndrome, denoted by $\rho(\langle u, v\rangle_w)$. When the testing vertex w is non-faulty, a result of $\rho(\langle u, v\rangle_w) = 0$ implies that both u and v are non-faulty; a result of $\rho(\langle u, v\rangle_w) = 1$ indicates that at least one of u and v is faulty. When the tester w is faulty, the test yields an unpredictable result: $\rho(\langle u, v\rangle_w) \in \{0, 1\}$ regardless of the statuses of u and v.

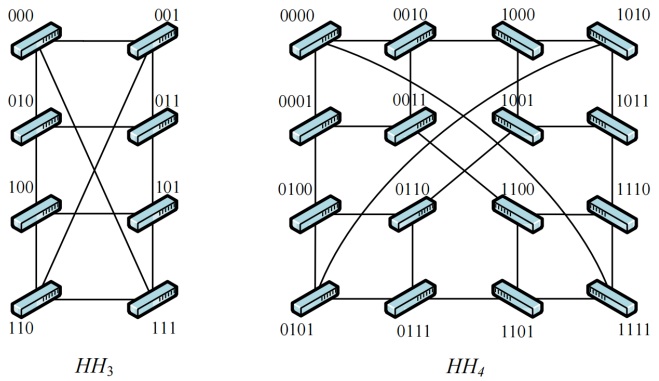


Fig. 1 The topologies of HH_3 and HH_4.

Theorem 1. [6] Let $G = (V(G), E(G))$ be a multiprocessor system. Under the MM* model, two distinct subsets $\mathcal{Z}_1, \mathcal{Z}_2 \subseteq V(G)$ are distinguishable iff they satisfy one or more of the following conditions.

(1) $|N_G(\{x, y\}) \cap (\mathcal{Z}_1 \Delta \mathcal{Z}_2)| \geq 1$, where $(x, y) \in E(\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2})$;

(2) $|N_G(y) \cap (\mathcal{Z}_i \setminus \mathcal{Z}_j)| \geq 2$, where $i, j \in \{1, 2\}$, $i \neq j$, and $y \in V(\overline{\mathcal{Z}_i \cup \mathcal{Z}_j})$.

The following definitions present the concepts of h-extra connectivity, h-extra r-component connectivity and diagnosability.

Definition 1. [12], [30] Suppose $G = (V(G), E(G))$ is a connected graph.

(1) Consider a vertex subset $X \subseteq V(G)$. If $G - X$ is disconnected and each of its components has order at least $h + 1$, then X constitutes an h-extra cut. The h-extra connectivity $\kappa_h(G)$ is given by

$$\kappa_h(G) = \min\{|X| : X \text{ is an } h\text{-extra cut of } G\}.$$

(2) We say that a vertex subset $X \subseteq V(G)$ constitutes an h-extra r-component cut whenever the removal of X from G leaves a disconnected subgraph and satisfies the following two conditions: (i) $G - X$ has at least r connected components; (ii) every component of $G - X$ has order at least $h + 1$. The h-extra r-component connectivity of G, denoted by $ck_r^h(G)$, is defined as the minimum cardinality among all such h-extra r-component cuts, i.e.,

$$ck_r^h(G) = \min\{|X| : X \text{ is an } h\text{-extra } r\text{-component cut of } G\}.$$

Definition 2. [32] A graph G is said to be h-extra r-component t-diagnosable iff for any two distinct h-extra r-component cuts $\mathcal{Z}_1, \mathcal{Z}_2 \subseteq V(G)$ with $max\{|\mathcal{Z}_1|, |\mathcal{Z}_2|\} \leq t$, the pair $(\mathcal{Z}_1, \mathcal{Z}_2)$ is distinguishable. The h-extra r-component diagnosability of G, denoted by $ct_r^h(G)$, is defined as the maximum integer t such that G is h-extra r-component t-diagnosable.

2.2 The Half Hypercube Network HH_n

Definition 3. [35] Let HH_n denote the n-dimensional half hypercube. Its vertex set $V(HH_n)$ and edge set $E(HH_n)$ are defined as follows.

Formally, $V(HH_n)$ is defined as:

$$V(HH_n) = \{x_n x_{n-1} \cdots x_1 \mid x_i \in \{0,1\},\ i = 1,2,\ldots,n\}.$$

For a vertex $x \in V(HH_n)$, let $L(x)$ denote the substring formed by the leftmost $\lfloor \frac{n}{2} \rfloor$ bits of x, and let $R(x)$ denote the remaining bits. That is,

$$L(x) = x_n x_{n-1} \cdots x_{\lceil \frac{n}{2} \rceil + 1}, R(x) = x_{\lceil \frac{n}{2} \rceil} x_{\lceil \frac{n}{2} \rceil - 1} \cdots x_1.$$

Additionally, we define $R'(x)$ as follows:

$$R'(x) = \begin{cases} x_{\lceil \frac{n}{2} \rceil} x_{\lceil \frac{n}{2} \rceil - 1} \cdots x_2 x_1, & \text{if } n \text{ is even;} \\ x_{\lceil \frac{n}{2} \rceil} x_{\lceil \frac{n}{2} \rceil - 1} \cdots x_2, & \text{if } n \text{ is odd.} \end{cases}$$

We now proceed to define the edge set $E(HH_n)$. For any two vertices $x = x_n x_{n-1} \cdots x_1$ and $y = y_n y_{n-1} \cdots y_1$ in $V(HH_n)$, an edge $(x, y) \in E(HH_n)$ iff it satisfies exactly one of the following two conditions:

(1) $E(HH_n) = \{(x, y) \mid L(x) = L(y), \sum_{i=1}^{\lceil \frac{n}{2} \rceil} |x_i - y_i| = 1\}$,

(2) $E(HH_n) = \{(x, y) \mid$ if $L(x) \neq R'(x)$, then $L(y) = R'(x), R'(y) = L(x)$; otherwise, $y = \bar{x}\}$, where $\bar{x}$ represents the binary complement of the sequence x.

The structure of HH_n is illustrated in Fig. 1, where $n = 3, 4$.

The following theorem presents some key properties of HH_n.

Theorem 2. [35], [36] The half hypercube HH_n has the following properties:

(1) HH_n is $(\lceil \frac{n}{2} \rceil + 1)$-regular and $|V(HH_n)| = 2^n$.

(2) HH_n is triangle-free.

(3) For any two distinct vertices $x, y \in V(HH_n)$ with $n \geq 3$, the number of common neighbors of x and y is at most 2.

(4) HH_n can be partitioned into $2^{\lfloor n/2 \rfloor}$ subcubes $HH_n^0, HH_n^1, \ldots, HH_n^{2^{\lfloor n/2 \rfloor}-1}$, each isomorphic to $Q_{\lceil n/2 \rceil}$, where each subcube consists of vertices with the same $L(x)$.

(5) Each vertex in HH_n has only one external neighbor.

(6) For $n \geq 6$ and any two distinct subcubes HH_n^i and HH_n^j, every vertex $v \in V(HH_n^j)$ has at most one neighbor within HH_n^j whose external neighbor belongs to HH_n^i.

3 Theoretical Results

This section establishes the 1-extra $\{2, 3\}$-component diagnosability of HH_n under the MM* model. Prior to this, we introduce several Lemmas and Theorems that are crucial for the derivation.

Lemma 1. [7] Let $F \subseteq V(HH_n)$ with $|F| \leq -\frac{1}{2}g^2 + (\lceil \frac{n}{2} \rceil - \frac{1}{2})g + \lceil \frac{n}{2} \rceil$, where $n \geq 7$ and $0 \leq g \leq \lceil \frac{n}{2} \rceil - 4$. Then $HH_n - F$ is either connected, or it consists of one large component together with several small components whose total number of vertices does not exceed g.

Lemma 2. [7] Let $S \subseteq V(HH_n)$ with $|S| = g$, where $n \geq 7$ and $1 \leq g \leq \lceil \frac{n}{2} \rceil + 1$. Then $|N_{HH_n}(S)| \geq g\lceil \frac{n}{2} \rceil - \frac{g(g-1)}{2} + 1$.

Theorem 3. [10] The 1-extra 3-component connectivity of HH_n satisfies $c\kappa_3^1(HH_n) = 4\lceil \frac{n}{2} \rceil - 4$ for $n \geq 7$.

Theorem 4. [37] A graph G possesses a perfect matching precisely when $o(G - S) \leq |S|$ for every $S \subseteq V(G)$, where $o(G - S)$ is the number of odd order components in $G - S$.

Lemma 3. [38] For any integer $n \geq 4$ and any vertex subset $S \subseteq V(Q_n)$ of size $|S| = g$, $|N_{Q_n}(S)| \geq gn - \frac{g(g+1)}{2} + 1$.

Lemma 4. [39] Let $n \geq 3$. In the hypercube Q_n:
(1) Let $U = \{t_1, t_2, t_3, t_4\}$. If $\{t_1, t_2, t_3\}$ forms a P_3 and $N_{Q_n}(\{t_1, t_2, t_3\}) \cap \{t_4\} = \emptyset$, then $|N_{Q_n}(\{t_1, t_2, t_3\}) \cap N_{Q_n}(\{t_4\})| \leq 3$.
(2) Let $U = \{t_1, t_2, t_3, t_4, t_5\}$. If (t_1, t_2), (t_2, t_3), and (t_4, t_5) are edges, and $N_{Q_n}(\{t_4, t_5\}) \cap \{t_1, t_2, t_3\} = \emptyset$, then $|N_{Q_n}(U)| \geq 5n - 11$.

Theorem 5. For $n \geq 7$, let $U = \{t_1, t_2, t_3, t_4, t_5\} \subseteq V(HH_n)$ such that (t_1, t_2), (t_2, t_3), and $(t_4, t_5) \in E(HH_n)$, and $E[\{t_1, t_2, t_3\}, \{t_4, t_5\}] = \emptyset$. Then, $|N_{HH_n}(U)| \geq 5\lceil \frac{n}{2} \rceil - 6$.

Proof. By Theorem 2 (4) and (5), each subcube $HH_n^i \cong Q_{\lceil n/2 \rceil}$ and every vertex has exactly one external neighbor. Next, we consider the following scenarios. based on the distribution of the vertices of U across the subcubes.

Scenario 1: All vertices of U are contained in one subcube.

Let $U \subseteq V(HH_n^i)$. By Lemma 4 (2), $|N_{HH_n^i}(U)| \geq 5\lceil \frac{n}{2} \rceil - 11$. By Theorem 2 (5), U has five distinct external neighbors. Thus, $|N_{HH_n}(U)| = |N_{HH_n^i}(U)| + 5 \geq 5\lceil \frac{n}{2} \rceil - 6$.

Scenario 2: The vertices of U are contained in two subcubes. Let $I = U \cap V(HH_n^i), J = U \cap V(HH_n^j)$, where $i \neq j$. W.l.o.g., assume that $|I| \geq |J|$. Then $(|I|, |J|) \in \{(4, 1), (3, 2)\}$.

Scenario 2.1: $|I| = 4$ and $|J| = 1$.

By Theorem 2 (5), $HH_n^i[I]$ consists of either two disjoint edges or a P_3 and an isolated vertex.

Suppose that $I = A \cup B$, where A and B are the vertex sets of the two edges. By the bipartiteness of HH_n^i and Theorem 2 (3), it follows that $|N_{HH_n^i}(A) \cap N_{HH_n^i}(B)| \leq 4$. Thus,

$$\begin{aligned} |N_{HH_n^i}(I)| &= |N_{HH_n^i}(A)| + |N_{HH_n^i}(B)| - |N_{HH_n^i}(A) \cap N_{HH_n^i}(B)| \\ &\geq 2\left(2\left\lceil \frac{n}{2} \right\rceil - 2\right) - 4 \\ &= 4\left\lceil \frac{n}{2} \right\rceil - 8. \end{aligned}$$

Suppose that $I = A \cup \{x\}$, where $HH_n^i[A] \cong P_3$ and x is isolated in $HH_n^i[I]$. By Lemmas 3 and 4 (1), we have

$$\begin{aligned} |N_{HH_n^i}(I)| &= |N_{HH_n^i}(A)| + |N_{HH_n^i}(x)| - |N_{HH_n^i}(A) \cap N_{HH_n^i}(x)| \\ &\geq \left(3\left\lceil \frac{n}{2} \right\rceil - 5\right) + \left\lceil \frac{n}{2} \right\rceil - 3 \\ &= 4\left\lceil \frac{n}{2} \right\rceil - 8. \end{aligned}$$

Hence, in both cases, $|N_{HH_n^i}(I)| \geq 4\lceil n/2 \rceil - 8$. Let $J = \{v\}$. Since $HH_n[U]$ has no isolated vertices, v has exactly one neighbor in I by Theorem 2 (5). Therefore, I has three distinct external neighbors outside U, while v has no external neighbor outside U. By Theorem 2 (6), v has at most one neighbor within HH_n^j whose external neighbor

belongs to HH_n^i. Since $I \subseteq V(HH_n^i)$, at most one of the three external neighbors of I outside U belongs to $N_{HH_n^j}(v)$. Consequently,

$$\begin{aligned}|N_{HH_n}(U)| &\geq |N_{HH_n^i}(I)| + |N_{HH_n^j}(J)| + 3 - 1\\ &\geq \left(4\left\lceil\frac{n}{2}\right\rceil - 8\right) + \left\lceil\frac{n}{2}\right\rceil + 2\\ &= 5\left\lceil\frac{n}{2}\right\rceil - 6.\end{aligned}$$

Scenario 2.2: $|I| = 3$ and $|J| = 2$.

If $HH_n^i[I] \cong P_3$, then $HH_n^j[J] \cong P_2$ and $E[I, J] = \emptyset$. By Lemma 3, $|N_{HH_n^i}(I)| \geq 3\lceil\frac{n}{2}\rceil - 5$ and $|N_{HH_n^j}(J)| \geq 2\lceil\frac{n}{2}\rceil - 2$. By Theorem 2 (5) and (6), U has five distinct external neighbors outside U, and each vertex of J has at most one neighbor within HH_n^j whose external neighbor belongs to HH_n^i. Since $|J| = 2$ and $I \subseteq V(HH_n^i)$, at most two external neighbors of I belong to $N_{HH_n^j}(J)$. Moreover, J has only two external neighbors, so at most two of them belong to $N_{HH_n^i}(I)$. Therefore,

$$\begin{aligned}|N_{HH_n}(U)| &\geq |N_{HH_n^i}(I)| + |N_{HH_n^j}(J)| + 5 - 4\\ &\geq \left(3\left\lceil\frac{n}{2}\right\rceil - 5\right) + \left(2\left\lceil\frac{n}{2}\right\rceil - 2\right) + 1\\ &= 5\left\lceil\frac{n}{2}\right\rceil - 6.\end{aligned}$$

For the other case, combining Theorem 2 (5) with the structure of $HH_n[U]$, we conclude that $HH_n^i[I]$ consists of an edge and an isolated vertex. Let

$$I = \{x, y, z\}, (x, y) \in E(HH_n^i), E[\{z\}, \{x, y\}] = \emptyset.$$

By the bipartiteness of HH_n^i, z cannot have common neighbors with both x and y. Thus, by Theorem 2 (3), we have $|N_{HH_n^i}(z) \cap N_{HH_n^i}(\{x, y\})| \leq 2$. It follows that

$$\begin{aligned}|N_{HH_n^i}(I)| &= |N_{HH_n^i}(\{x, y\})| + |N_{HH_n^i}(z)|\\ &\quad - |N_{HH_n^i}(\{x, y\}) \cap N_{HH_n^i}(z)|\\ &\geq \left(2\left\lceil\frac{n}{2}\right\rceil - 2\right) + \left\lceil\frac{n}{2}\right\rceil - 2\\ &= 3\left\lceil\frac{n}{2}\right\rceil - 4.\end{aligned}$$

By Lemma 3, $|N_{HH_n^j}(J)| \geq 2\lceil\frac{n}{2}\rceil - 2$. Since $N_{HH_n^i}(I)$ and $N_{HH_n^j}(J)$ are disjoint subsets of $N_{HH_n}(U)$, we obtain

$$\begin{aligned}|N_{HH_n}(U)| &\geq |N_{HH_n^i}(I)| + |N_{HH_n^j}(J)|\\ &\geq \left(3\left\lceil\frac{n}{2}\right\rceil - 4\right) + \left(2\left\lceil\frac{n}{2}\right\rceil - 2\right)\\ &= 5\left\lceil\frac{n}{2}\right\rceil - 6.\end{aligned}$$

Scenario 3: The vertices of U are contained in at least three subcubes.

By Theorem 2 (5), at least one of t_1 and t_3 belongs to the subcube containing t_2. Thus, U intersects at most four subcubes, with nonempty intersection sizes $(3, 1, 1)$, $(2, 2, 1)$, or $(2, 1, 1, 1)$. By Lemma 3, for $X \subseteq V(HH_n^i)$ with $1 \leq |X| \leq 3$,

$$|N_{HH_n^i}(X)| \geq \begin{cases}\lceil\frac{n}{2}\rceil, & |X| = 1,\\ 2\lceil\frac{n}{2}\rceil - 2, & |X| = 2,\\ 3\lceil\frac{n}{2}\rceil - 5, & |X| = 3.\end{cases}$$

Therefore,

$$\begin{aligned}|N_{HH_n}(U)| &\geq \min\left\{\left(3\left\lceil\frac{n}{2}\right\rceil - 5\right) + 2\left\lceil\frac{n}{2}\right\rceil,\ 2\left(2\left\lceil\frac{n}{2}\right\rceil - 2\right) + \left\lceil\frac{n}{2}\right\rceil,\right.\\ &\qquad \left.\left(2\left\lceil\frac{n}{2}\right\rceil - 2\right) + 3\left\lceil\frac{n}{2}\right\rceil\right\}\\ &= 5\left\lceil\frac{n}{2}\right\rceil - 5\\ &> 5\left\lceil\frac{n}{2}\right\rceil - 6.\end{aligned}$$

Combining all scenarios, we obtain

$$|N_{HH_n}(U)| \geq 5\left\lceil\frac{n}{2}\right\rceil - 6.$$

□

Theorem 6. For $n \geq 13$, $ct_2^1(HH_n) \leq 3\lceil\frac{n}{2}\rceil - 2$ under the MM* model.

Proof. The following argument establishes an upper bound for $ct_2^1(HH_n)$. Let $U = \{u_1, u_2, u_3\}$ be a vertex subset of HH_n satisfying $(u_1, u_2), (u_2, u_3) \in E(HH_n)$. Consider two vertex subsets $\mathcal{Z}_1 = N_{HH_n}(V(U)) \cup \{u_1\}$ and $\mathcal{Z}_2 = N_{HH_n}(V(U)) \cup \{u_3\}$. Obviously, $\mathcal{Z}_1 \Delta \mathcal{Z}_2 = \{u_1, u_3\}$ and $\mathcal{Z}_1 \cap \mathcal{Z}_2 = N_{HH_n}(V(U))$. Thus, $(\mathcal{Z}_1, \mathcal{Z}_2)$ is an indistinguishable pair under the MM* model by Theorem 1. According to Theorem 2, u_1 and u_3 have another neighbor in $\mathcal{Z}_1 \cap \mathcal{Z}_2$ (see Fig. 2(a)), it can be derive $|N_{HH_n}(V(U))| = 3(\lceil\frac{n}{2}\rceil + 1 - 2) + 1 = 3(\lceil\frac{n}{2}\rceil) - 2$. So $|\mathcal{Z}_1| = |\mathcal{Z}_2| = 3(\lceil\frac{n}{2}\rceil) - 1$. Note that

$$\begin{aligned}|\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2} \setminus \{u_2\}| &= |V(HH_n)| - |\mathcal{Z}_1 \cup \mathcal{Z}_2| - |\{u_2\}|\\ &\geq |V(HH_n)| - (|\mathcal{Z}_1| + |\mathcal{Z}_2|) - |\{u_2\}|\\ &\geq 2^n - 2(3\lceil\frac{n}{2}\rceil - 1) - 1\\ &\geq 2^n - 6\lceil\frac{n}{2}\rceil + 1\\ &> 2.\end{aligned}$$

This implies that there exists at least one additional component in $\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2} \setminus \{u_2\}$. Then, $HH_n - \mathcal{Z}_1$ (or $HH_n - \mathcal{Z}_2$) contains not less than two components. In the following, we demonstrate that the degree of each vertex is at least one in $\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2} \setminus \{u_2\}$. To derive a contradiction, suppose that $t \in \overline{\mathcal{Z}_1 \cup \mathcal{Z}_2} \setminus \{u_2\}$ is an isolated vertex, if $N_{HH_n}(t) \cap \{u_1, u_2, u_3\} \neq \emptyset$, we obtain $t \in N_{HH_n}(V(U))$, leading to a contradiction. Thus, $N_{HH_n}(t) \subseteq \mathcal{Z}_1 \cap \mathcal{Z}_2$. At shown in Fig. 2(b), t shares at most two common neighbors with u_1, u_2, and u_3 based on Theorem 2, respectively. It follows that $|\mathcal{Z}_1 \cap \mathcal{Z}_2| \geq 2(\lceil\frac{n}{2}\rceil + 1 - 3) + (\lceil\frac{n}{2}\rceil + 1 - 4) + (\lceil\frac{n}{2}\rceil + 1 - 6) + 6 = 4\lceil\frac{n}{2}\rceil - 6$, and then $|\mathcal{Z}_1| = |\mathcal{Z}_1 \cap \mathcal{Z}_2| + |\mathcal{Z}_1 \setminus \mathcal{Z}_2| = 4\lceil\frac{n}{2}\rceil - 5 > 3\lceil\frac{n}{2}\rceil - 1 \geq |\mathcal{Z}_1|$ for $n \geq 13$, contradicting the assumption. Hence, each vertex has at least one neighbor in $\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2} \setminus \{u_2\}$. Then $\mathcal{Z}_1$ and $\mathcal{Z}_2$ are two distinct

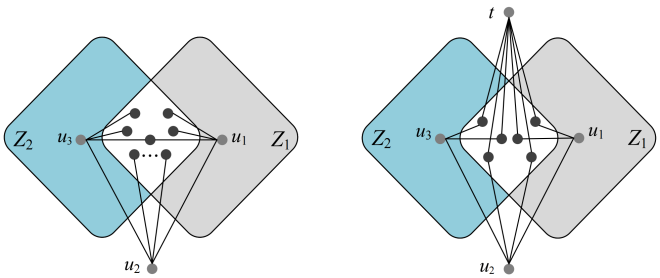


Fig. 2 Illustration of Theorem 6.

1-extra 2-component vertex cuts, and they are indistinguishable under the MM* model. Therefore, $ct_2^1(HH_n) \leq 3\lceil\frac{n}{2}\rceil - 2$ as desired. □

Theorem 7. For $n \geq 13$, $ct_2^1(HH_n) \geq 3\lceil\frac{n}{2}\rceil - 2$ under the MM* model.

Proof. We proceed to derive a lower bound for $ct_2^1(HH_n)$. Assume that $ct_2^1(HH_n) \leq 3\lceil\frac{n}{2}\rceil - 3$. Take $\mathcal{Z}_1$ and $\mathcal{Z}_2$ as two distinct 1-extra 2-component vertex cuts satisfying $max\{|\mathcal{Z}_1|, |\mathcal{Z}_2|\} \leq 3\lceil\frac{n}{2}\rceil - 2$ and they are indistinguishable under the MM* model. When $n \geq 13$,

$$\begin{aligned}|\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}| &= |V(HH_n)| - |\mathcal{Z}_1 \cup \mathcal{Z}_2| \\ &= |V(HH_n)| - (|\mathcal{Z}_1| + |\mathcal{Z}_2| - |\mathcal{Z}_1 \cap \mathcal{Z}_2|) \\ &\geq 2^n - 2(3\lceil\frac{n}{2}\rceil - 2) \\ &> 4.\end{aligned}$$

Hence, $V(HH_n) \neq \mathcal{Z}_1 \cup \mathcal{Z}_2$. Next, one claim that $\mathcal{Z}_1 \setminus \mathcal{Z}_2 \neq \emptyset$ and $\mathcal{Z}_2 \setminus \mathcal{Z}_1 \neq \emptyset$. If $\mathcal{Z}_1 \setminus \mathcal{Z}_2 = \emptyset$, i.e., $\mathcal{Z}_1 \subseteq \mathcal{Z}_2$. Since both of $\mathcal{Z}_1$ and $\mathcal{Z}_2$ satisfy the definition of 1-extra 2-component cut, $V(HH_n) - \mathcal{Z}_2$ contains at least two components, each with at least two vertices, and $\mathcal{Z}_2 \setminus \mathcal{Z}_1 \neq \emptyset$ as well as $|\mathcal{Z}_2 \setminus \mathcal{Z}_1| \geq 2$. Thus, $\mathcal{Z}_1$ is a 1-extra 3-component cut. As stated in Theorem 3, $|\mathcal{Z}_1| \geq c\kappa_3^1(HH_n) = 4\lceil\frac{n}{2}\rceil - 4$. Therefore,

$$|\mathcal{Z}_2| = |\mathcal{Z}_1| + |\mathcal{Z}_2 \setminus \mathcal{Z}_1| \geq 4\lceil\frac{n}{2}\rceil - 2 > 3\lceil\frac{n}{2}\rceil - 2 \geq |\mathcal{Z}_2|,$$

a contradiction. Hence, $\mathcal{Z}_1 \setminus \mathcal{Z}_2 \neq \emptyset$ and $\mathcal{Z}_2 \setminus \mathcal{Z}_1 \neq \emptyset$ hold. This indicates that $|\mathcal{Z}_i \setminus \mathcal{Z}_j| \geq 1$ for $i \neq j$ and $i, j \in \{1, 2\}$.

According to $|\mathcal{Z}_1 \setminus \mathcal{Z}_2| \geq 1$, we have $|\mathcal{Z}_1 \cap \mathcal{Z}_2| \leq 3\lceil\frac{n}{2}\rceil - 3$. When $g = 3$ and $n \geq 13$, applying Lemma 1, $3\lceil\frac{n}{2}\rceil - 3 \leq 4\lceil\frac{n}{2}\rceil - 6$ for $n \geq 13$. Thus, $HH_n - (\mathcal{Z}_1 \cap \mathcal{Z}_2)$ has a large component of size at least $|V(HH_n)| - |\mathcal{Z}_1 \cap \mathcal{Z}_2| - 3$. Let H represent the union of the small components in $HH_n - (\mathcal{Z}_1 \cap \mathcal{Z}_2)$ with $|H| \leq 3$. Now consider whether an isolated vertex exists in $\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}$. Let X be the set of all isolated vertices in $\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}$. Due to the fact that $\mathcal{Z}_1$ and $\mathcal{Z}_2$ are 1-extra 2-component cuts with indistinguishable under the MM* model, we have $|N_{HH_n}(t) \cap (\mathcal{Z}_1 \setminus \mathcal{Z}_2)| = |N_{HH_n}(t) \cap (\mathcal{Z}_2 \setminus \mathcal{Z}_1)| = 1$ for $t \in X$. This implies $|(\mathcal{Z}_1 \Delta \mathcal{Z}_2) \cup \{t\}| \geq 3$. Clearly, $(\mathcal{Z}_1 \Delta \mathcal{Z}_2) \cup \{t\}$ is one small component of $HH_n - (\mathcal{Z}_1 \cap \mathcal{Z}_2)$, i.e., $H = (\mathcal{Z}_1 \Delta \mathcal{Z}_2) \cup \{t\}$. Thus, $|(\mathcal{Z}_1 \Delta \mathcal{Z}_2) \cup \{t\}| = 3$ and $N_{HH_n}((\mathcal{Z}_1 \Delta \mathcal{Z}_2) \cup \{t\}) \subseteq \mathcal{Z}_1 \cap \mathcal{Z}_2$. Without loss of generality, take $x \in \mathcal{Z}_1 \setminus \mathcal{Z}_2$ and $y \in \mathcal{Z}_2 \setminus \mathcal{Z}_1$ with $(x, t), (t, y) \in E(HH_n)$. Then, by Theorem 2, there exists an additional neighbor of both x and y in $\mathcal{Z}_1 \cap \mathcal{Z}_2$. Thus, we derive

$$\begin{aligned}|\mathcal{Z}_1 \cap \mathcal{Z}_2| &\geq |N_{HH_n}((\mathcal{Z}_1 \Delta \mathcal{Z}_2) \cup \{t\})| \\ &\geq 3(\lceil\frac{n}{2}\rceil + 1 - 2) + 1 \\ &= 3\lceil\frac{n}{2}\rceil - 2,\end{aligned}$$

and so

$$|\mathcal{Z}_1| = |\mathcal{Z}_1 \cap \mathcal{Z}_2| + |\mathcal{Z}_1 \setminus \mathcal{Z}_2| \geq 3\lceil\frac{n}{2}\rceil - 1 > |\mathcal{Z}_1|.$$

This leads to a contradiction. Hence, $\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}$ contains no isolated vertices.

Given that there exists no isolated vertex in $\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}$, for $t \in \overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}$, there exists at least one vertex $w \in \overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}$ such that $(t, w) \in E(\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2})$. By Theorem 1, $\mathcal{Z}_1$ and $\mathcal{Z}_2$ are indistinguishable under the MM* model, we have $E[\mathcal{Z}_1 \Delta \mathcal{Z}_2, \overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}] = \emptyset$. Meanwhile, since $\mathcal{Z}_1$ and $\mathcal{Z}_2$ are 1-extra 2-component cuts, $|\mathcal{Z}_1 \setminus \mathcal{Z}_2| \geq 2$, $|\mathcal{Z}_2 \setminus \mathcal{Z}_1| \geq 2$, and then $|\mathcal{Z}_1 \Delta \mathcal{Z}_2| \geq 4$. Remarking that $V(\mathcal{Z}_1 \Delta \mathcal{Z}_2) = V(HH_n) - \mathcal{Z}_1 \cap \mathcal{Z}_2 - \overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}$ and $E[\mathcal{Z}_1 \Delta \mathcal{Z}_2, \overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}] = \emptyset$, it implies that $N_{HH_n}(\mathcal{Z}_1 \Delta \mathcal{Z}_2) \subseteq \mathcal{Z}_1 \cap \mathcal{Z}_2$. By Lemma 1 with $g = 3$, $HH_n - (\mathcal{Z}_1 \cap \mathcal{Z}_2)$ consists of one large component $\mathcal{T}$ and small components whose total number of vertices at most 3. Since $|\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}| > 4$ and $E[\mathcal{Z}_1 \Delta \mathcal{Z}_2, \overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}] = \emptyset$, we have $V(\mathcal{T}) \subseteq \overline{\mathcal{Z}_1 \cup \mathcal{Z}_2}$. Hence,

$$|\mathcal{Z}_1 \Delta \mathcal{Z}_2| \leq \left|V(HH_n) \setminus \left((\mathcal{Z}_1 \cap \mathcal{Z}_2) \cup V(\mathcal{T})\right)\right| \leq 3.$$

This contradicts $|\mathcal{Z}_1 \Delta \mathcal{Z}_2| \geq 4$. Hence, $ct_2^1(HH_n) \geq 3\lceil n/2\rceil - 2$ for $n \geq 13$. □

Combining Theorems 6 and 7, we determine $ct_2^1(HH_n) = 3\lceil\frac{n}{2}\rceil - 2$ for $n \geq 13$ under the MM* model.

The following Theorems 8 and 9 establish the upper bound and lower bound of $ct_3^1(HH_n)$, respectively.

Theorem 8. For $n \geq 19$, $ct_3^1(HH_n) \leq 5\lceil\frac{n}{2}\rceil - 6$ under the MM* model.

Proof. Let $U = U_1 \cup U_2$ be a vertex set of HH_n with $U_1 = \{u_1, u_2, u_3\}$ and $U_2 = \{u_4, u_5\}$. W.l.o.g, consider $u_1 = 0^{n-4}0000$, $u_2 = 0^{n-4}0001$, $u_3 = 0^{n-4}1001$, $u_4 = 0^{n-4}0110$, $u_5 = 0^{n-4}0111$. Then, (u_1, u_2), (u_2, u_3), and $(u_4, u_5) \in E(HH_n)$. Let $\mathcal{Z}_1 = N_{HH_n}(V(U)) \cup \{u_1\}$ and $\mathcal{Z}_2 = N_{HH_n}(V(U)) \cup \{u_3\}$. Observe that $\mathcal{Z}_1 \Delta \mathcal{Z}_2 = \{u_1, u_3\}$ and $\mathcal{Z}_1 \cap \mathcal{Z}_2 = N_{HH_n}(V(U))$. Therefore, $\mathcal{Z}_1$ and $\mathcal{Z}_2$ are indistinguishable under the MM* model. As depicted in Fig. 3(a), u_1 and u_4 share at most two common neighbors $c_1 = 0^{n-4}0100$ and $c_2 = 0^{n-4}0010$ by Theorem 2; similarly, u_2 and u_5 share up to two common neighbors $c_3 = 0^{n-4}0101$ and $c_4 = 0^{n-4}0011$; u_1 and u_3 have another neighbor $c_5 = 0^{n-4}1000$. It can be calculated that $|N_{HH_n}(V(U))| = 2(\lceil\frac{n}{2}\rceil + 1 - 4) + (\lceil\frac{n}{2}\rceil + 1 - 2) + 2(\lceil\frac{n}{2}\rceil + 1 - 3) + 5 = 5\lceil\frac{n}{2}\rceil - 6$, and so

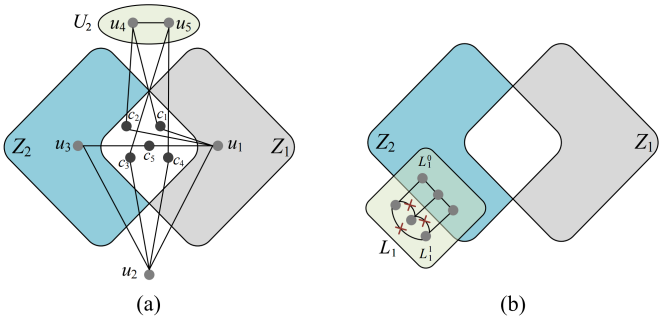


Fig. 3 Illustration of Theorems 8 and 9.

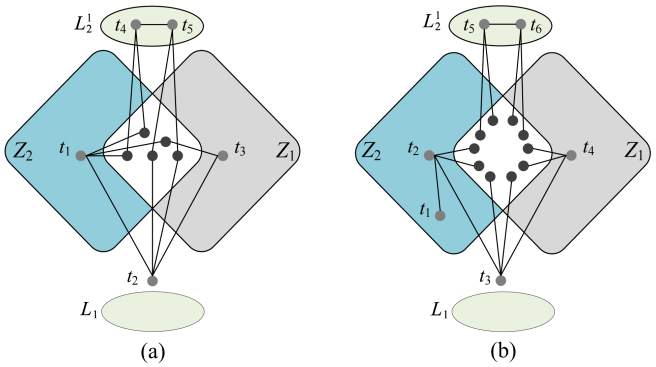


Fig. 4 Illustration for Scenarios 1 and 2 of Theorem 9.

$|\mathcal{Z}_1| = |\mathcal{Z}_2| = 5\lceil\frac{n}{2}\rceil - 5$. Let $M = \overline{\mathcal{Z}_1 \cup \mathcal{Z}_2} \setminus \{u_2, u_4, u_5\}$, we have

$$\begin{aligned}|M| &= |\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2} \setminus \{u_2, u_4, u_5\}| \\ &= |V(HH_n)| - |\mathcal{Z}_1 \cup \mathcal{Z}_2| - 3 \\ &\geq |V(HH_n)| - (|\mathcal{Z}_1| + |\mathcal{Z}_2|) - 3 \\ &\geq 2^n - 2(5\lceil\frac{n}{2}\rceil - 5) - 3 \\ &\geq 2^n - 10\lceil\frac{n}{2}\rceil + 7 \\ &> 2.\end{aligned}$$

Hence, removing $\mathcal{Z}_1$ (or $\mathcal{Z}_2$) from HH_n yields at least three components, two of which are the edges (u_2, u_3) (or (u_1, u_2)) and (u_4, u_5).

Next, we prove that M contains no isolated vertices by contradiction. For any vertex $t \in M$, we have $E[M, U] = \emptyset$ and $N_{HH_n}(V(M)) \subseteq \mathcal{Z}_1 \cap \mathcal{Z}_2$; otherwise, $t \in N_{HH_n}(V(U))$, a contradiction. As stated in Theorem 2, t shares at most two common neighbors with each $u_i \in U$, implying that $|N_{HH_n}(t) \cap (\bigcup_{u_i \in U} N_{HH_n}(u_i))| \leq 10$. Since HH_n is $(\lceil\frac{n}{2}\rceil + 1)$-regular, t has additional neighbors in M for $n \geq 19$. Therefore, M contains no isolated vertices. So $\mathcal{Z}_1$ and $\mathcal{Z}_2$ are two distinct 1-extra 3-components cuts, and they are indistinguishable under the MM* model. Hence, $ct_3^1(HH_n) \leq 5\lceil\frac{n}{2}\rceil - 6$. □

Theorem 9. For $n \geq 19$, $ct_3^1(HH_n) \geq 5\lceil\frac{n}{2}\rceil - 6$ under the MM* model.

Proof. For contradiction, assume that $ct_3^1(HH_n) \leq 5\lceil\frac{n}{2}\rceil - 7$. Consider two distinct 1-extra 3-component cuts, denoted $\mathcal{Z}_1$ and $\mathcal{Z}_2$, satisfying $max\{|\mathcal{Z}_1|, |\mathcal{Z}_2|\} \leq 5\lceil\frac{n}{2}\rceil - 6$ and they are indistinguishable under the MM* model. Clearly,

$$\begin{aligned}\overline{\mathcal{Z}_1 \cup \mathcal{Z}_2} &= |V(HH_n)| - |\mathcal{Z}_1 \cup \mathcal{Z}_2| \\ &= |V(HH_n)| - (|\mathcal{Z}_1| + |\mathcal{Z}_2| - |\mathcal{Z}_1 \cap \mathcal{Z}_2|) \\ &\geq 2^n - 2(5\lceil\frac{n}{2}\rceil - 6) \\ &> 4.\end{aligned}$$

Thus, $V(HH_n) \neq \mathcal{Z}_1 \cup \mathcal{Z}_2$. Based on Lemma 1, when $g = 5$ and $n \geq 19$, $5\lceil\frac{n}{2}\rceil - 6 \leq 6\lceil\frac{n}{2}\rceil - 15$. It implies that $HH_n - \mathcal{Z}_1$ contains one large component, denoted by L_1, with $|L_1| \geq 2^n - |\mathcal{Z}_1| - 5$. Let L_2 be the union of the remaining small components in $HH_n - \mathcal{Z}_1$ with $|L_2| \leq 5$. Moreover, L_2 consists of at least two components, denoted by L_2^0 and L_2^1, where $|L_2^0| \geq |L_2^1|$. Consequently, we obtain $2 \leq |L_2^0| \leq 3$ and $|L_2^1| = 2$. Recall that $|L_1| \geq 2^n - |\mathcal{Z}_1| - 5$, and we have

$$\begin{aligned}|V(L_1)| - |\mathcal{Z}_2| &\geq 2^n - |\mathcal{Z}_1| - 5 - |\mathcal{Z}_2| \\ &\geq 2^n - 10\lceil\frac{n}{2}\rceil + 7 \\ &> 0.\end{aligned}$$

Hence, $L_1 \nsubseteq \mathcal{Z}_2 \setminus \mathcal{Z}_1$.

Now, we show that $V(L_1) \cap (\mathcal{Z}_2 \setminus \mathcal{Z}_1) = \emptyset$ holds. Assume, by contradiction, that $V(L_1) \cap (\mathcal{Z}_2 \setminus \mathcal{Z}_1) \neq \emptyset$. Since $|V(L_1)| - |\mathcal{Z}_2| > 0$, $V(L_1) \setminus (\mathcal{Z}_2 \setminus \mathcal{Z}_1) \neq \emptyset$. Consider $L_1^0 = V(L_1) \cap (\mathcal{Z}_2 \setminus \mathcal{Z}_1)$ and $L_1^1 = V(L_1) \setminus (\mathcal{Z}_2 \setminus \mathcal{Z}_1)$. Due to the fact that L_1 is a connected component, we have $E[L_1^0, L_1^1] \neq \emptyset$. In terms of Theorem 1, L_1^1 forms an independent set in $HH_n - (\mathcal{Z}_1 \cup \mathcal{Z}_2)$ (see Fig. 3(b)); otherwise, $\mathcal{Z}_1$ and $\mathcal{Z}_2$ are distinguishable under the MM* model, a contradiction with the assumption. By Definition 3, the edge set $\mathcal{P} = \{(x_n \cdots x_2 0, x_n \cdots x_2 1) \mid x_n, \ldots, x_2 \in \{0, 1\}\}$ is a perfect matching of HH_n. Indeed, the two vertices of each edge in $\mathcal{P}$ differ only in the last bit, and every vertex of HH_n belongs to exactly one edge in $\mathcal{P}$. By Theorem 4, $|L_1^1| \leq o\big(HH_n - (\mathcal{Z}_1 \cup \mathcal{Z}_2)\big) \leq |\mathcal{Z}_1 \cup \mathcal{Z}_2| \leq |\mathcal{Z}_1| + |\mathcal{Z}_2|$.

Since $L_1 = L_1^0 \cup L_1^1$ and $|L_1^0| \leq |\mathcal{Z}_2 \setminus \mathcal{Z}_1|$, we obtain

$$\begin{aligned}|V(L_1)| &= |V(L_1^0)| + |V(L_1^1)| \\ &\leq |\mathcal{Z}_2 \setminus \mathcal{Z}_1| + |\mathcal{Z}_1| + |\mathcal{Z}_2| \\ &\leq |\mathcal{Z}_1| + 2|\mathcal{Z}_2| \\ &\leq 15\lceil\frac{n}{2}\rceil - 18.\end{aligned}$$

This contradicts $|V(L_1)| \geq 2^n - |\mathcal{Z}_1| - 5 = 2^n - 5\lceil\frac{n}{2}\rceil + 1$ for $n \geq 19$. Thus, $V(L_1) \cap (\mathcal{Z}_2 \setminus \mathcal{Z}_1) = \emptyset$. By the same token, we have $V(L_1) \cap (\mathcal{Z}_1 \setminus \mathcal{Z}_2) = \emptyset$. So L_1 is also one large component of $HH_n - (\mathcal{Z}_1 \cap \mathcal{Z}_2)$.

Next, given $2 \leq |V(L_2^0)| \leq 3$ and $|V(L_2^1)| = 2$, we proceed to examine the following two scenarios.

Scenario 1: $|V(L_2^0)| = 2$ and $|V(L_2^1)| = 2$.

In this scenario, both L_2^0 and L_2^1 are one 2-paths P_2 in $HH_n - \mathcal{Z}_1$. Without loss of generality, consider $(t_1, t_2) \in L_2^0$ and $(t_4, t_5) \in L_2^1$, where $t_1 \in (\mathcal{Z}_2 \setminus \mathcal{Z}_1)$ and $t_2 \in \overline{(\mathcal{Z}_1 \cup \mathcal{Z}_2)}$. Given that $(\mathcal{Z}_1, \mathcal{Z}_2)$ is an indistinguishable pair under the MM* model, and

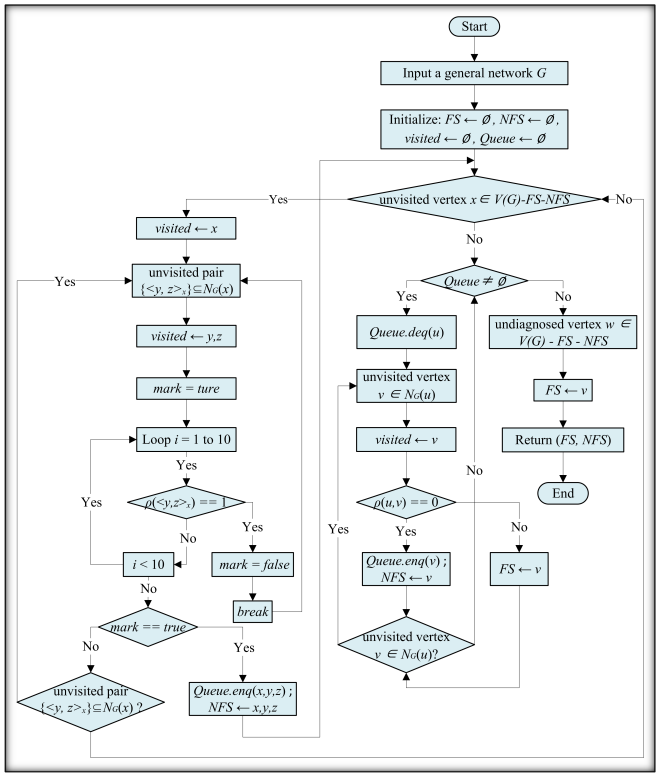


Fig. 5 The flowchart of Algorithm 1.

both $\mathcal{Z}_1$ and $\mathcal{Z}_2$ satisfy the 1-extra condition, we have $|N_{HH_n}(t_2) \cap (\mathcal{Z}_1 \setminus \mathcal{Z}_2)| = |N_{HH_n}(t_2) \cap (\mathcal{Z}_2 \setminus \mathcal{Z}_1)| = 1$. Thus, we obtain $(t_2, t_3) \in E(HH_n)$ for $t_3 \in (\mathcal{Z}_1 \setminus \mathcal{Z}_2)$ (see Fig. 4(a)). Since $V(HH_n - \mathcal{Z}_1 \cap \mathcal{Z}_2) = \{t_1, t_2, t_3, t_4, t_5\} \cup V(L_1)$, which indicates that $N_{HH_n}(\{t_1, t_2, t_3, t_4, t_5\}) \subseteq \mathcal{Z}_1 \cap \mathcal{Z}_2$. Therefore, $|\mathcal{Z}_1 \cap \mathcal{Z}_2| \geq |N_{HH_n}(\{t_1, t_2, t_3, t_4, t_5\})| \geq 5\lceil \frac{n}{2} \rceil - 6$ based on Theorem 5. Hence, $|\mathcal{Z}_1| = |\mathcal{Z}_1 \setminus \mathcal{Z}_2| + |\mathcal{Z}_1 \cap \mathcal{Z}_2| \geq 5\lceil \frac{n}{2} \rceil - 5 > |\mathcal{Z}_1|$, a contradiction.

Scenario 2: $|V(L_2^0)| = 3$ and $|V(L_2^1)| = 2$.

Obviously, L_2^0 is one 3-path P_3 and L_2^1 is one 2-path P_2. Let (t_1, t_2), $(t_2, t_3) \in L_2^0$, and $(t_5, t_6) \in L_2^1$. If $t_1 \in (\mathcal{Z}_2 \setminus \mathcal{Z}_1)$, $t_2 \in \overline{(\mathcal{Z}_1 \cup \mathcal{Z}_2)}$, and $t_3 \in (\mathcal{Z}_1 \setminus \mathcal{Z}_2)$, the case is identical to the situation described in Scenario 1. If $\{t_1, t_2\} \in (\mathcal{Z}_2 \setminus \mathcal{Z}_1)$, $t_3 \in \overline{(\mathcal{Z}_1 \cup \mathcal{Z}_2)}$. Analogous to the analysis in Scenario 1, we have $|N_{HH_n}(t_3) \cap (\mathcal{Z}_1 \setminus \mathcal{Z}_2)| = |N_{HH_n}(t_3) \cap (\mathcal{Z}_2 \setminus \mathcal{Z}_1)| = 1$, $(t_3, t_4) \in E(HH_n)$ for $t_4 \in (\mathcal{Z}_1 \setminus \mathcal{Z}_2)$ (see Fig. 4(b)), and $V(HH_n - \mathcal{Z}_1 \cap \mathcal{Z}_2) = \{t_1, t_2, t_3, t_4, t_5, t_6\} \cup V(L_1)$. So $N_{HH_n}(\{t_1, t_2, t_3, t_4, t_5, t_6\}) \subseteq \mathcal{Z}_1 \cap \mathcal{Z}_2$. Applying Lemma 2, $|\mathcal{Z}_1 \cap \mathcal{Z}_2| \geq |N_{HH_n}(\{t_1, t_2, t_3, t_4, t_5, t_6\})| \geq 6\lceil \frac{n}{2} \rceil - 14$. Thus, $|\mathcal{Z}_1| = |\mathcal{Z}_1 \setminus \mathcal{Z}_2| + |\mathcal{Z}_1 \cap \mathcal{Z}_2| \geq 6\lceil \frac{n}{2} \rceil - 13 > 5\lceil \frac{n}{2} \rceil - 6 \geq |\mathcal{Z}_1|$ for $n \geq 19$, a contradiction. Therefore, $ct_3^1(HH_n) \geq 5\lceil \frac{n}{2} \rceil - 6$. □

From Theorems 8 and 9, we derive $ct_3^1(HH_n) = 5\lceil \frac{n}{2} \rceil - 6$ for $n \geq 19$ under the MM* model.

4 Algorithm and Experiment Analysis

In this section, we design a dual-model fusion diagnosis algorithm, termed $DMFDA$, which integrates the MM* and PMC diagnostic frameworks to accurately identify faulty vertices in general networks.

Algorithm 1: Dual-Model Fusion Diagnosis Algorithm Based on the MM* and PMC Models

Input: A general network G.

Output: The faulty set FS and the non-faulty set NFS.

Initialize: $FS \leftarrow \emptyset$, $NFS \leftarrow \emptyset$, $visited \leftarrow \emptyset$, $Queue \leftarrow \emptyset$;
for *unvisited vertex* $x \in V(G) - FS - NFS$ **do**
  $visited \leftarrow x$; // Record x as visited and add it to the set $visited$
  for *selected disjoint neighbor pair* $\langle y, z \rangle_x$ *with* $y, z \in N_G(x) \setminus visited$ **do**
    // y and z are a pair of neighbors of x in G
    $visited \leftarrow y, z$;
    $mark = true$;
    for $i = 1$ ***to*** 10 **do**
      if $\rho(\langle y, z \rangle_x) == 1$ **then**
        $mark = false$;
        $break$; // Terminate the current loop and proceed to the next pair of neighbors
      end
    end
    if $mark == true$ **then**
      $Queue.enq(x, y, z)$; // Enqueue the non-faulty vertices x, y, z to $Queue$
      $NFS \leftarrow x, y, z$; // Add the non-faulty vertices x, y, z to NFS
    end
  end
end
while $Queue \neq \emptyset$ **do**
  $Queue.deq(u)$; // Dequeue the non-faulty vertex u from $Queue$
  for *undiagnosed vertex* $v \in N_G(u) \setminus (FS \cup NFS)$ **do**
    $visited \leftarrow v$;
    if $\rho(u, v) == 0$ **then**
      $Queue.enq(v)$;
      $NFS \leftarrow v$;
    end
    else
      $FS \leftarrow v$; // Add the faulty vertex v to FS
    end
  end
end
for *undiagnosed vertex* $w \in V(G) - FS - NFS$ **do**
  $FS \leftarrow w$;
end
return (FS, NFS);

The flowchart of Algorithm 1 is presented in Fig. 5. Simulation experiments on HH_n evaluate the algorithm under two fault patterns. We also conduct ablation studies and sensitivity analysis on HH_{13}, followed by a comparison with existing diagnosis algorithms through fault-injection simulations on the $DCell_{(2,4)}$ topology.

4.1 Dual-Model Fusion Diagnosis Algorithm

We now briefly outline Algorithm 1. This algorithm combines the MM* and PMC diagnostic frameworks to locate faulty units in general networks. Specifically, it selects initial diagnostic seeds through repeated MM* comparisons of selected neighbor pairs. Repeated zero outcomes serve as a probabilistic acceptance criterion under the random output setting. The accepted seeds then initiate the PMC diagnosis phase. The detailed steps of the $DMFDA$ algorithm are presented below.

Step 1: Set up the faulty set FS, the non-faulty set NFS, the visited set $visited$, and the queue $Queue$ as empty. The set $visited$ records the visited nodes, and the queue $Queue$ temporarily stores the identified non-faulty vertices.

Steps 2-18: For each unvisited vertex x in G, we first add x to $visited$, then traverse all its neighboring vertices and pair them up. We illustrate how to pair the neighbors of a vertex x using an example. If x has an odd number of neighbors, e.g., $N_G(x) = \{a, b, c, d, y, z, k\}$. Then it can be partitioned into three pairs, for instance $\{\langle a, b\rangle_x, \langle c, d\rangle_x, \langle y, z\rangle_x\} \subseteq N_G(x)$, or $\{\langle a, c\rangle_x, \langle b, d\rangle_x, \langle y, k\rangle_x\} \subseteq N_G(x)$, with the remaining vertex excluded from pairing. For each selected unvisited neighbor pair $\langle y, z\rangle_x$, we add y and z to $visited$ and initialize $mark = true$. Vertex x performs at most ten comparisons on this pair. If any comparison returns 1, we set $mark = false$, stop the repeated comparisons for the current pair, and examine the next selected neighbor pair of x. If all ten outcomes are 0, we accept x, y, and z as non-faulty seeds, enqueue them into $Queue$, and add them to NFS. The algorithm then examines the remaining selected neighbor pairs of x. After all selected pairs have been examined, it proceeds to the next vertex.

Steps 19-31: While $Queue$ is not empty, the following operations are performed. A vertex u classified as non-faulty is dequeued from $Queue$. For each undiagnosed neighbor $v \in N_G(u) \setminus (FS \cup NFS)$, we add v to $visited$ and determine its status using the PMC test result $\rho(u, v)$. If $\rho(u, v) == 0$, v is identified as a non-faulty vertex, enqueued into $Queue$, and added to NFS. Otherwise, v is classified as faulty and placed into the fault set FS.

Steps 32-34: After completing the diagnostic steps described above, we uniformly classify any remaining undiagnosed vertices in G into FS.

Step 35: Return the complete sets FS and NFS.

Theorem 10. Let F be the actual faulty vertex set of G, and let S_0 be the initial seed set accepted before the PMC diagnosis phase of Algorithm 1. $DMFDA$ correctly classifies all vertices if and only if $S_0 \subseteq V(G) \setminus F$, and every connected component C of $G - F$ satisfies $V(C) \cap S_0 \neq \emptyset$.

Proof. Suppose that both conditions hold. Then all initial seeds are non-faulty. Under the PMC test rules, a non-faulty tester correctly identifies the status of each tested neighbor. Each newly identified non-faulty vertex then serves as a tester. Since every connected component of $G - F$ contains an initial seed, PMC propagation identifies all non-faulty vertices. All remaining undiagnosed vertices are faulty. Therefore, the algorithm returns $NFS = V(G) \setminus F$, $FS = F$.

If $S_0 \cap F \neq \emptyset$, at least one faulty vertex is accepted as a non-faulty seed and remains in NFS. Thus, the final classification is incorrect. Suppose that $S_0 \cap F = \emptyset$, but a connected component of $G - F$ contains no initial seed. The non-faulty vertices in this component cannot be identified through PMC propagation. The algorithm then adds them to FS in the final step, which produces false positives. Therefore, both conditions are necessary for correct classification of all vertices. □

Diagnosis errors arise from faulty seed selection and the final classification of undiagnosed vertices. In the simulations, a faulty tester or comparator independently returns 0 or 1 with equal probability in each PMC test or MM* comparison. During seed selection, a comparator and its selected neighbor pair are accepted as non-faulty seeds if all ten comparisons return 0. For a faulty comparator, the probability of this event for a selected pair is $(1/2)^{10} = 2^{-10}$. Once accepted as a seed, the faulty comparator performs PMC tests, and an incorrect test outcome leads to the misclassification of the tested neighbor. After PMC propagation, the algorithm assigns all remaining undiagnosed vertices to FS. This step correctly classifies undiagnosed faulty vertices, while assigning undiagnosed non-faulty vertices to FS produces false positives.

Theorem 11. The time complexity of the $DMFDA$ algorithm is $O(N\Delta)$, where N indicates the total amount of vertices in G and Δ denotes the maximum degree of vertices in G.

Proof. We analyze the time complexity of the proposed $DMFDA$ algorithm by breaking down its execution into distinct steps, as detailed in Algorithm 1.

Step 1: The sets FS, NFS, $visited$, and the queue $Queue$ can be initialized in $O(1)$ time.

Steps 2-18:

For a vertex x, the algorithm examines at most $\lfloor |N_G(x)|/2 \rfloor$ selected neighbor pairs. Each pair requires at most ten comparisons. Consequently, the time complexity for processing a single node x is $O(\Delta)$. Summing over all nodes in G, the total time complexity of this phase is $O(N\Delta)$.

Steps 19-31: Each non-faulty vertex is dequeued at most once, resulting in $O(N)$ dequeue operations in the worst case. For each dequeued vertex u, we traverse its neighbors (at most Δ). Thus, the total time complexity of this phase is $O(N\Delta)$.

Steps 32-34: The traversal and classification of all remaining undiagnosed vertices in G into FS require linear time $O(N)$ in the worst case.

Step 35: The complete sets FS and NFS are returned in $O(1)$ time.

Thus, the overall time complexity of the $DMFDA$ algorithm is $O(N\Delta)$. □

Table 3 Evaluation metrics.

Indicator	Description
TP	The count of faulty vertices that are correctly identified as faulty
TN	The count of non-faulty vertices that are correctly identified as non-faulty
FP	The count of non-faulty vertices that are mistakenly classified as faulty
FN	The count of faulty vertices that are mistakenly classified as non-faulty
$ACCR$	The proportion of correct diagnoses among all test cases
TPR	The rate at which actual faulty vertices are successfully detected
MCC	A balanced measure that accounts for all four categories (TP, TN, FP, FN) to assess classification quality
FPR	The rate at which non-faulty vertices are erroneously flagged as faulty

4.2 Simulation Experiments

To systematically evaluate the diagnostic performance of the $DMFDA$ algorithm, we conduct simulation experiments on the half hypercube network HH_n under two 1-extra r-component fault patterns, namely $r = 2$ and $r = 3$. The experiments cover dimensions $n \in \{10, 11, 12, 13\}$ and fault rates ranging from 10% to 50%. For each network dimension and fault rate p, the number of faulty vertices is set to $\lceil pN \rceil$, where $N = 2^n$ is the number of vertices in HH_n. Faulty vertices are randomly selected subject to the corresponding 1-extra r-component condition. When a faulty vertex acts as a tester, we independently generate its outcome for each test as 0 or 1, each with probability 0.5. This setting applies to both PMC tests and MM* comparisons. For each configuration, we perform 100 independent trials and report the arithmetic mean of each evaluation metric across these trials. Four metrics are employed to quantify the algorithm's effectiveness: Accuracy ($ACCR$), True Positive Rate (TPR), Matthews Correlation Coefficient (MCC), and False Positive Rate (FPR). The interpretation of each metric is summarized in Table 3. The specific formulas are given below:

$$
\begin{aligned}
ACCR &= \frac{TP + TN}{TP + TN + FP + FN}, \\
TPR &= \frac{TP}{TP + FN}, \\
MCC &= \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \\
FPR &= \frac{FP}{FP + TN}.
\end{aligned}
$$

The simulation experiments are conducted on a computer equipped with a 13th Gen Intel Core i5-13500 processor running at 2.50 GHz and 16.0 GB of RAM. The computer ran a 64-bit Windows operating system on an x64-based processor. The experiments are implemented in Python 3.13 using PyCharm, with NumPy 2.5.1 and NetworkX 3.6.1 as the main libraries.

Table 4 Summary of evaluation metrics of HH_n under 1-extra 2-component fault pattern.

Indicator	p	$n = 10$	$n = 11$	$n = 12$	$n = 13$
$ACCR$	10%	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
	20%	0.9977±0.0004	0.9996±0.0002	1.0000±0.0000	1.0000±0.0000
	30%	0.9964±0.0008	0.9974±0.0006	0.9984±0.0004	0.9994±0.0002
	40%	0.9932±0.0015	0.9944±0.0012	0.9949±0.0010	0.9957±0.0008
	50%	0.9834±0.0025	0.9888±0.0020	0.9890±0.0018	0.9909±0.0015
TPR	10%	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
	20%	0.9985±0.0005	0.9995±0.0003	1.0000±0.0000	1.0000±0.0000
	30%	0.9951±0.0012	0.9966±0.0008	0.9993±0.0005	0.9998±0.0002
	40%	0.9930±0.0018	0.9957±0.0015	0.9969±0.0012	0.9978±0.0010
	50%	0.9921±0.0030	0.9930±0.0025	0.9935±0.0020	0.9947±0.0018
MCC	10%	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
	20%	0.9984±0.0006	0.9987±0.0004	1.0000±0.0000	1.0000±0.0000
	30%	0.9946±0.0015	0.9971±0.0010	0.9974±0.0006	0.9995±0.0003
	40%	0.9858±0.0025	0.9912±0.0018	0.9921±0.0015	0.9949±0.0012
	50%	0.9768±0.0040	0.9888±0.0030	0.9890±0.0025	0.9929±0.0020
FPR	10%	0.0000±0.0000	0.0000±0.0000	0.0000±0.0000	0.0000±0.0000
	20%	0.0004±0.0001	0.0000±0.0000	0.0000±0.0000	0.0000±0.0000
	30%	0.0031±0.0005	0.0002±0.0001	0.0001±0.0001	0.0000±0.0000
	40%	0.0054±0.0008	0.0030±0.0005	0.0017±0.0004	0.0011±0.0003
	50%	0.0153±0.0015	0.0131±0.0012	0.0126±0.0010	0.0064±0.0008

We first examine the performance of the $DMFDA$ algorithm under the 1-extra 2-component fault pattern. As shown in Fig. 6 and Table 4, at a 10% fault rate, the reported mean values of $ACCR$, TPR, and MCC are 1, while the mean FPR is 0 across all tested dimensions. At a 20% fault rate, the algorithm maintains high diagnostic performance. For example, when $n = 10$, $ACCR$ is 0.9977 and FPR is 0.0004. As the fault rate increases from 30% to 50%, $ACCR$, TPR, and MCC gradually decrease, while FPR increases. When $n = 13$ and the fault rate is 50%, $ACCR$ is 0.9909, TPR is 0.9947, and MCC is 0.9929, while FPR is 0.0064. The results also show changes in diagnostic performance across network dimensions. For example, at a 30% fault rate, as n increases from 10 to 13, $ACCR$ increases from 0.9964 to 0.9994, TPR increases from 0.9951 to 0.9998, and MCC increases from 0.9946 to 0.9995, while FPR decreases from 0.0031 to 0.

We next examine the performance of the $DMFDA$ algorithm under the 1-extra 3-component fault pattern. As shown in Fig. 7 and Table 5, at fault rates of 10% and 20%, the reported mean values of $ACCR$, TPR, and MCC are 1.0000, while the mean FPR is 0 across all tested dimensions. At a 30% fault rate, the algorithm continues to achieve high diagnostic performance. For example, when $n = 10$, $ACCR$ is 0.9968, TPR is 0.9987, and MCC is 0.9985, while FPR is 0.0021. As the fault rate increases to 40% and 50%, $ACCR$, TPR, and MCC decrease, while FPR increases. When $n = 13$ and the fault rate is 50%, $ACCR$ is 0.9961, TPR is 0.9965, and MCC is 0.9968, while FPR is 0.0026. At the same fault rate, when n increases from 10 to 13, $ACCR$

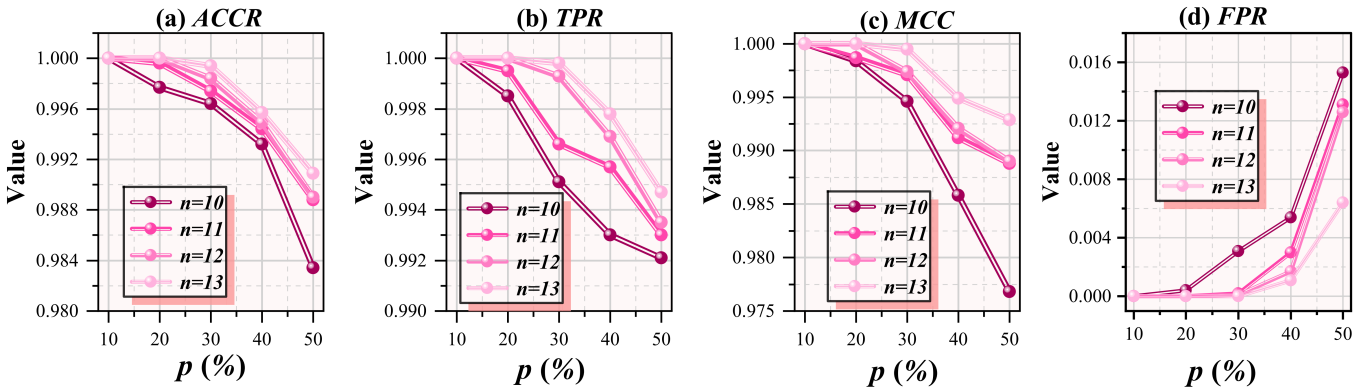


Fig. 6 Experiment results of HH_n under the 1-extra 2-component fault pattern.

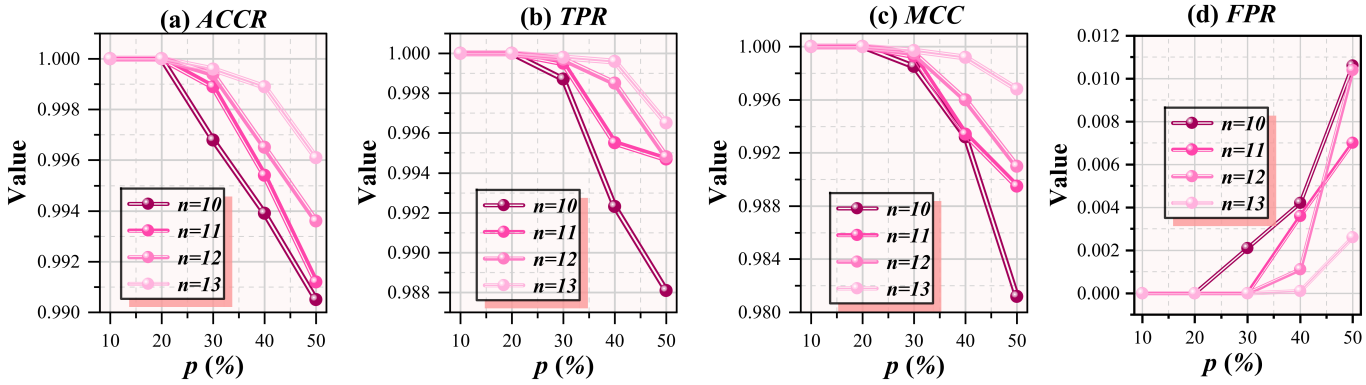


Fig. 7 Experiment results of HH_n under the 1-extra 3-component fault pattern.

Table 5 Summary of evaluation metrics of HH_n under 1-extra 3-component fault pattern.

Indicator	p	n = 10	n = 11	n = 12	n = 13
	10%	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
	20%	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
$ACCR$	30%	0.9968±0.0012	0.9989±0.0006	0.9994±0.0003	0.9996±0.0002
	40%	0.9939±0.0020	0.9954±0.0012	0.9965±0.0008	0.9989±0.0005
	50%	0.9905±0.0035	0.9912±0.0022	0.9936±0.0015	0.9961±0.0010
	10%	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
	20%	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
TPR	30%	0.9987±0.0008	0.9995±0.0004	0.9997±0.0002	0.9998±0.0001
	40%	0.9923±0.0015	0.9955±0.0010	0.9985±0.0006	0.9996±0.0003
	50%	0.9881±0.0025	0.9947±0.0018	0.9948±0.0012	0.9965±0.0008
	10%	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
	20%	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
MCC	30%	0.9985±0.0010	0.9993±0.0005	0.9995±0.0003	0.9997±0.0002
	40%	0.9932±0.0018	0.9934±0.0012	0.9960±0.0008	0.9992±0.0005
	50%	0.9812±0.0030	0.9895±0.0020	0.9910±0.0015	0.9968±0.0010
	10%	0.0000±0.0000	0.0000±0.0000	0.0000±0.0000	0.0000±0.0000
	20%	0.0000±0.0000	0.0000±0.0000	0.0000±0.0000	0.0000±0.0000
FPR	30%	0.0021±0.0004	0.0000±0.0000	0.0000±0.0000	0.0000±0.0000
	40%	0.0042±0.0008	0.0036±0.0006	0.0011±0.0003	0.0001±0.0001
	50%	0.0106±0.0015	0.0070±0.0010	0.0104±0.0012	0.0026±0.0005

increases from 0.9905 to 0.9961, TPR increases from 0.9881 to 0.9965, and MCC increases from 0.9812 to 0.9968, while FPR changes from 0.0106 to 0.0026.

4.3 Ablation Studies and Comparative Experiments

In this subsection, we evaluate the performance of $DMFDA$ through ablation studies, sensitivity analysis, and comparisons with existing diagnosis algorithms. The ablation studies and sensitivity analysis are conducted on HH_{13}. The comparison with existing algorithms is conducted on the data center network $DCell_{(2,4)}$. The evaluation uses a static $DCell_{(2,4)}$ topology, with fault sets and diagnostic outcomes generated in the simulation. For the $DCell_{(2,4)}$ experiments, the fault rates are set to 30% and 50%, and fault sets are randomly generated subject to the 1-extra 2-component condition. Diagnostic outcomes are generated according to the PMC and MM* model rules. A non-faulty tester produces outcomes according to the corresponding diagnostic model, whereas a faulty tester independently returns 0 or 1 with equal probability for each test or comparison. Each experimental configuration is evaluated through 100 independent trials. The total time reported in Tables 6 and 7 is the sum of the diagnosis times over the 100 trials for each configuration. In each trial, time is measured from the start of diagnosis to the completion of vertex classification.

To evaluate the performance of combining the two diagnostic models, Table 6 compares $DMFDA$ with the PMC-only and MM*-only baselines on HH_{13} at fault rates of 10% and 50%. At the 10% fault rate, the mean $ACCR$ and TPR of $DMFDA$ are both 1.0000, compared

Table 6 Ablation study on the HH_{13}: performance and time overhead comparison against single-model baselines under 10% and 50% fault rates.

p	Method	$ACCR$	TPR	Total Time (s)
10%	PMC-only	0.9991 (±0.0004)	0.9911 (±0.0041)	23.4
	MM*-only	0.9993 (±0.0003)	0.9942 (±0.0038)	13.2
	$DMFDA$	**1.0000 (±0.0000)**	**1.0000 (±0.0000)**	**11.0**
50%	PMC-only	0.9883 (±0.0024)	0.9766 (±0.0048)	14.6
	MM*-only	0.9833 (±0.0017)	0.9838 (±0.0043)	8.7
	$DMFDA$	**0.9909 (±0.0015)**	**0.9947 (±0.0018)**	**6.4**

Table 7 Sensitivity analysis of the number of repetitions evaluated at HH_{13}.

Repetitions	p	$ACCR$	TPR	Total Time (s)
3	10%	0.9509±0.0019	0.5103±0.0194	5.3
	50%	0.6269±0.0048	0.2604±0.0092	4.7
5	10%	0.9853±0.0013	0.8525±0.0128	7.4
	50%	0.8476±0.0067	0.7055±0.0131	5.3
7	10%	0.990±0.0008	0.9604±0.0084	9.0
	50%	0.9420±0.0056	0.8883±0.0110	5.5
10	10%	**1.0000±0.0000**	**1.0000±0.0000**	11.0
	50%	**0.9909±0.0015**	**0.9947±0.0018**	6.4

Table 8 Performance comparison of the proposed algorithm with other algorithms at 30% and 50% fault rates on the $DCell_{(2,4)}$.

Algorithm	p	$ACCR$	TPR	MCC	FPR
ICFD-P [40]	30%	0.9714±0.0012	0.9376±0.0032	0.9328±0.0028	**0.0137±0.0006**
	50%	0.8842±0.0025	0.8337±0.0037	0.7735±0.0045	**0.0649±0.0025**
ACP-FD [41]	30%	0.9784±0.0010	0.9860±0.0016	0.9497±0.0024	0.0248±0.0013
	50%	0.9230±0.0021	0.9336±0.0028	0.8462±0.0042	0.0876±0.0030
$DMFDA$ **[Ours]**	30%	**0.9807±0.0011**	**0.9953±0.0014**	**0.9556±0.0025**	0.0255±0.0015
	50%	**0.9329±0.0019**	**0.9925±0.0016**	**0.8720±0.0035**	0.1267±0.0035

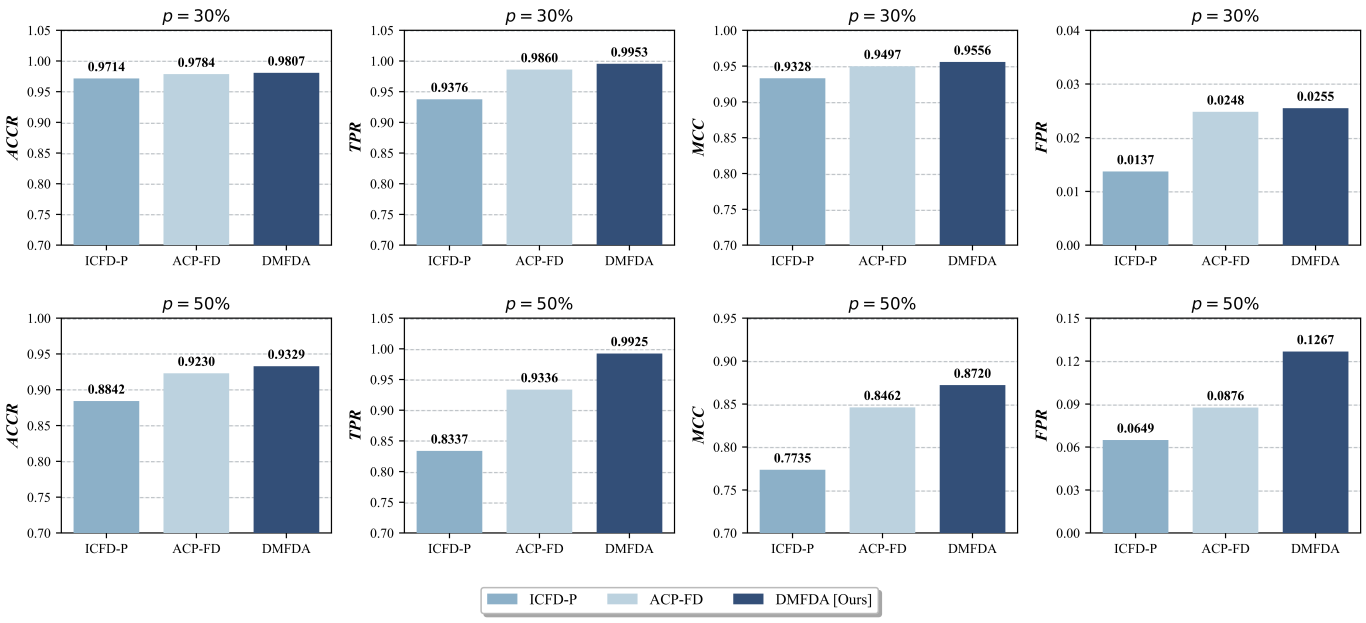


Fig. 8 Comparison of diagnostic performance of ICFD-P, ACP-FD, and $DMFDA$ on the $DCell_{(2,4)}$.

with 0.9991 and 0.9911 for PMC-only and 0.9993 and 0.9942 for MM*-only, respectively. At the 50% fault rate, the mean $ACCR$ and TPR of $DMFDA$ are 0.9909 and 0.9947, compared with 0.9883 and 0.9766 for PMC-only and 0.9833 and 0.9838 for MM*-only, respectively. The reported total time of $DMFDA$ is 11.0 s at the 10% fault rate and 6.4 s at the 50% fault rate. The corresponding times are 23.4 s and 14.6 s for PMC-only, and 13.2 s and 8.7 s for MM*-only. In these tested configurations, $DMFDA$ achieves higher mean $ACCR$ and TPR values with a shorter reported total time than the two single-model baselines.

Table 7 shows the effect of the number of repeated MM* comparisons on diagnostic performance and time cost. At the 10% fault rate, increasing the number of repetitions from 3 to 10 raises the mean TPR from 0.5103 to 1.0000 and the mean $ACCR$ from 0.9509 to 1.0000. At the 50% fault rate, the mean TPR increases from 0.2604 to 0.9947, and the mean $ACCR$ increases from 0.6269 to 0.9909. The reported total time increases from 5.3 s to 11.0 s at the 10% fault rate and from 4.7 s to 6.4 s at the 50% fault rate. Under the independent random-output

assumption for faulty testers described in Section 4.2, the probability that a faulty comparator returns zero in all k comparisons is 2^{-k}. Repeated comparisons therefore reduce the probability of accepting a faulty comparator based on repeated zero outcomes. The results show that increasing the number of repetitions improves the reported diagnostic performance while increasing the time cost.

To further evaluate $DMFDA$, we conduct fault diagnosis simulations on the data center network $DCell_{(2,4)}$, which contains 3263442 vertices. Table 8 and Fig. 8 compare $DMFDA$ with ICFD-P [40] and ACP-FD [41] at fault rates of 30% and 50%. At the 30% fault rate, $DMFDA$ achieves an $ACCR$ of 0.9807, a TPR of 0.9953, and an MCC of 0.9556. At the 50% fault rate, its $ACCR$, TPR, and MCC are 0.9329, 0.9925, and 0.8720, respectively. These values are higher than those of ICFD-P and ACP-FD at the corresponding fault rates. In terms of FPR, $DMFDA$ produces a value of 0.0255 at the 30% fault rate, compared with 0.0137 for ICFD-P and 0.0248 for ACP-FD. At the 50% fault rate, the FPR of $DMFDA$ is 0.1267, compared with 0.0649 for ICFD-P and 0.0876 for ACP-FD. These results show that $DMFDA$ identifies a larger proportion of faulty vertices in the tested configurations, while also classifying a larger proportion of fault-free vertices as faulty. The increase in false positives with the fault rate originates from two mechanisms in DMFDA. The first mechanism involves the MM* phase. A faulty comparator outputs 0 or 1 with a probability of 0.5. If a faulty vertex is accepted as an initial seed, it enters the PMC phase and misclassifies its correct neighbors. The second mechanism occurs in the final classification step. $DMFDA$ assigns vertices that remain undiagnosed after propagation to the faulty set. If these undiagnosed vertices are fault-free, this step also produces false positives.

5 Conclusion

This paper first conducts a rigorous theoretical analysis to determine the 1-extra r-component diagnosability of the half hypercube network HH_n under the MM* model and then presents the theoretical expressions of $ct_2^1(HH_n)$ and $ct_3^1(HH_n)$. Specifically, $ct_2^1(HH_n) = 3\lceil\frac{n}{2}\rceil - 2$ for $n \geq 13$ and $ct_3^1(HH_n) = 5\lceil\frac{n}{2}\rceil - 6$ for $n \geq 19$. Furthermore, we develop a dual-model fusion diagnosis algorithm, termed $DMFDA$, by combining the MM* and PMC diagnostic models. Simulation experiments are conducted on HH_n under the 1-extra 2-component and 1-extra 3-component fault patterns to evaluate the diagnostic performance of $DMFDA$. Ablation experiments with single-model baselines and sensitivity analysis of the number of repeated comparisons are also performed on HH_{13}. In addition, $DMFDA$ is compared with existing diagnosis algorithms on the data center network $DCell_{(2,4)}$, which contains 3263442 vertices. At a 50% fault rate on $DCell_{(2,4)}$, $DMFDA$ achieves a TPR of 0.9925 and an FPR of 0.1267. The results demonstrate the diagnostic performance of $DMFDA$ under the tested conditions and indicate that the detection of faulty vertices should be considered together with the misclassification of fault-free vertices.

Future work will focus on further optimizing the diagnostic workflow and memory consumption of the proposed algorithm. Furthermore, we will investigate multi conditional diagnosability in general networks and incorporate link faults into the diagnosis algorithm.

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Data availability

All data included in this study are available upon request to the corresponding author.

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